

Lecture 02

Evaluation Metrics – part 2

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Today's Class

- Evaluation Metrics for Recommender Systems
 - Evaluation paradigms: user studies, online evaluation, offline evaluation
 - Design issues in offline recommender evaluation
 - Accuracy metrics for offline evaluation
 - Limitations of evaluation measures



Accuracy Metrics for Offline Evaluation

Accuracy Metrics for Offline Evaluation

- Measuring the Accuracy of Ratings Prediction
 - Accuracy measures how well the model predicts a user's preference for items

Accuracy Metrics for Offline Evaluation

- Mean Square Error (MSE)

	v_1	v_2	v_3	...	v_m
u_1	2			3	?
u_2		?	5		5
u_3		?	3		
u_4	?				4
u_5		2		3	?
...	4		?		
			1	?	
		2			?
u_n	2			?	5

<User-Item Interaction Matrix>

Accuracy Metrics for Offline Evaluation

- **Mean Square Error (MSE)**

- Let D_{test} be a test data set
- Let $f(\cdot)$ be a recommendation model
- For each test entry $d = (u_i, v_j, r_{i,j})$ and $f(u_i, v_j) = \widehat{r}_{i,j}$ where $\widehat{r}_{i,j}$ is the predicted rating

$$MSE = \frac{\sum_{d \in D_{test}} (r_{i,j} - f(u_i, v_j))^2}{|D_{test}|} = \frac{\sum_{d \in D_{test}} (r_{i,j} - \widehat{r}_{i,j})^2}{|D_{test}|}$$

- Smaller values of the MSE are indicative of superior performance

Table 2. Ablation test results. SP and ET denote the second phase in our two-phase learning and the exponential transform in Equation 10, respectively. We observe the performance changes by removing each component.

Components			Entire Match		Warm-Warm Match		Warm-Cold Match		Cold-Cold Match	
e ^s	SP	ET	MSE (↓)	AUROC (↑)	MSE (↓)	AUROC (↑)	MSE (↓)	AUROC (↑)	MSE (↓)	AUROC (↑)
✗	✓	✓	0.4197	0.8497	0.3996	0.8655	0.4359	0.8375	0.4539	0.8136
✓	✗	✓	0.4271	0.8464	0.4059	0.8649	0.4436	0.8329	0.4648	0.7995
✓	✓	✗	0.3996	0.8615	0.3845	0.8712	0.4121	0.8545	0.4239	0.8394
✓	✓	✓	0.3948	0.8648	0.3797	0.8745	0.4072	0.8577	0.4190	0.8431

Accuracy Metrics for Offline Evaluation

- **Root Mean Square Error (RMSE) and Mean Absolute Error (MAE)**

$$RMSE = \sqrt{\frac{\sum_{d \in D_{test}} (r_{i,j} - f(u_i, v_j))^2}{|D_{test}|}} = \sqrt{\frac{\sum_{d \in D_{test}} (r_{i,j} - \widehat{r}_{i,j})^2}{|D_{test}|}}$$

$$MAE = \frac{\sum_{d \in D_{test}} |r_{i,j} - f(u_i, v_j)|}{|D_{test}|} = \frac{\sum_{d \in D_{test}} |r_{i,j} - \widehat{r}_{i,j}|}{|D_{test}|}$$

- As RMSE sums up the squared errors, it is disproportionately affected by large error values or outliers
- A few poorly predicted ratings can significantly deteriorate the RMSE measure

Accuracy Metrics for Offline Evaluation

- **Precision and Recall**

- Collaborative filtering can also be viewed as a generalization of **classification**
- Therefore, metrics for classification problems/tasks are also employed in the evaluation of recommender systems
- Precision and recall are popular metrics commonly used for implicit feedback data

	✓				✓	
		✓				✓
	?	✓	?	✓	?	?

<Implicit Feedback>

Accuracy Metrics for Offline Evaluation

- **Precision and Recall**

		Model Prediction	
		True	False
Actual Label	True	True Positive (TP)	False Negative (FN)
	False	False Positive (FP)	True Negative (TN)

$$\text{Precision} = \frac{TP}{TP+FP}$$

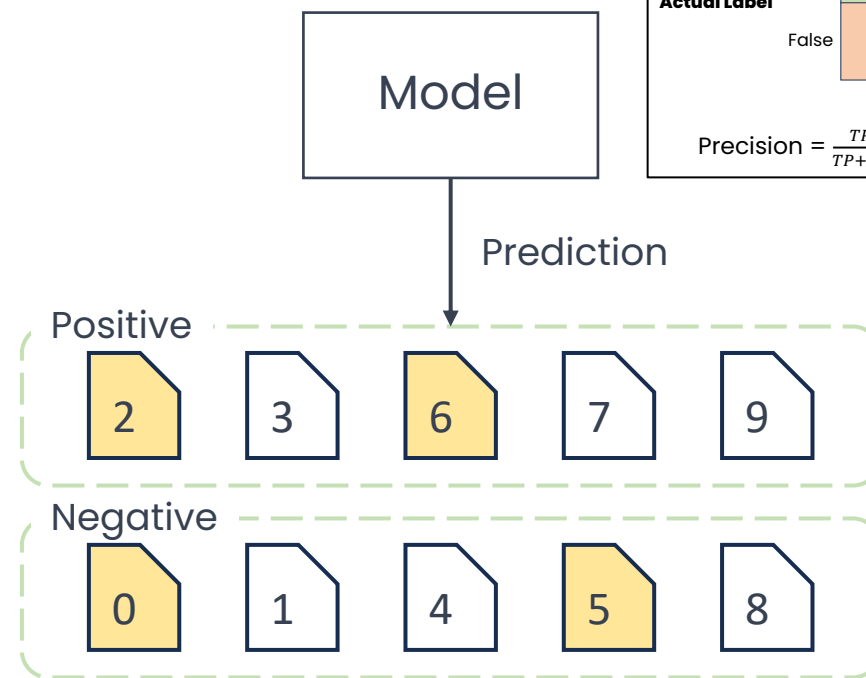
$$\text{Recall} = \frac{TP}{TP+FN}$$

Accuracy Metrics for Offline Evaluation

- Precision and Recall



$$\text{Precision} = \frac{(\text{2} \text{ } \text{6})}{(\text{2} \text{ } \text{3} \text{ } \text{6} \text{ } \text{7} \text{ } \text{9})} = 0.4$$



$$\text{Recall} = \frac{(\text{2} \text{ } \text{6})}{(\text{0} \text{ } \text{2} \text{ } \text{5} \text{ } \text{6})} = 0.5$$

		Model Prediction	
		True	False
Actual Label	True	True Positive (TP)	False Negative (FN)
	False	False Positive (FP)	True Negative (TN)

Precision = $\frac{TP}{TP+FP}$ Recall = $\frac{TP}{TP+FN}$

Accuracy Metrics for Offline Evaluation

- **Precision and Recall**

- Let U be a user set
- Let Y_u be a set of true interactions for a user $u \in U$
- Let $\widehat{Y}_u$ be a set of predicted true interactions by the recommendation model

$$precision = \frac{1}{|U|} \sum_{u \in U} \frac{|Y_u \cap \widehat{Y}_u|}{|\widehat{Y}_u|}, \quad recall = \frac{1}{|U|} \sum_{u \in U} \frac{|Y_u \cap \widehat{Y}_u|}{|Y_u|}$$

Accuracy Metrics for Offline Evaluation

- **F-Score**

- It is well-known that precision and recall have a trade-off relationship
- The F-Score integrates a model's precision and recall to provide a more comprehensive performance evaluation for users

$$F_1 \text{ score} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad F_\beta \text{ score} = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{(\beta^2 \cdot \text{precision}) + \text{recall}}$$

- The F_β -score uses a positive real factor β , where β is chosen to reflect the relative importance of recall in comparison to precision, with recall being considered β times as important as precision

Accuracy Metrics for Offline Evaluation

- **Area Under the Receiver Operating Characteristic Curve (AUROC, AUC)**
 - It is well-known that precision and recall have a trade-off relationship
 - The model's prediction are determined by a threshold τ

$$\hat{y} = \begin{cases} true, & f(x) \geq \tau \\ false, & f(x) < \tau \end{cases}$$

- As the threshold decreases, recall increases while precision decreases

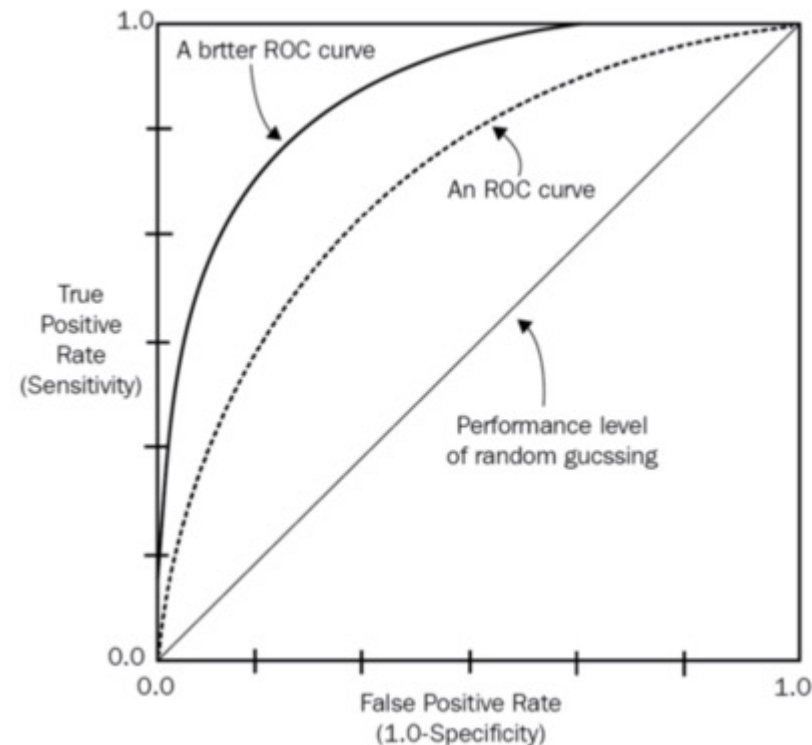
Accuracy Metrics for Offline Evaluation

- Area Under the Receiver Operating Characteristic Curve (AUROC, AUC)

$$\text{True Positive Rate (Recall)} = \frac{TP}{TP + FN}$$

$$\text{False Positive Rate} = \frac{FP}{FP + TN}$$

		Model Prediction	
		True	False
Actual Label	True	True Positive (TP)	False Negative (FN)
	False	False Positive (FP)	True Negative (TN)



Accuracy Metrics for Offline Evaluation

- **Area Under the Receiver Operating Characteristic Curve (AUROC, AUC)**
 - **Imbalanced Class Distribution**
 - In binary classification problems with imbalanced class distributions, evaluating model performance solely based on accuracy can be challenging
 - AUROC enhances the reliability of model performance assessment in cases of imbalanced class distributions
 - **Addressing Sensitivity to Thresholds**
 - Setting thresholds for classifying model predictions can impact model performance
 - AUROC provides an evaluation of the model's overall classification ability without being sensitive to the choice of threshold

Accuracy Metrics for Offline Evaluation

- Area Under the Receiver Operating Characteristic Curve (AUROC, AUC)

Table 1: Offline & online metrics of different models.
Online Acquisition Gain is relative to the control.

Model	Offline AUC	Online Acquisition Gain
Wide (control)	0.726	0%
Deep	0.722	+2.9%
Wide & Deep	0.728	+3.9%

Wide & Deep Learning for Recommender Systems (Google), Cheng, et al. 2016

Accuracy Metrics for Offline Evaluation

시네벤치R23(싱글): 1,666점 / 시네벤치R23(멀티): 8376 / 출시가: 97날라 (VAT별도)

관련기사 빠른 사후대체를 위해서는 정품 프로세서 구입은 필수, 인텔 프로세서는 어떻게 수입되는가?

사용기 ASRock B760M Pro RS D5 디앤디컴 + INTEL i3-12100F + DDR5 5600 업그레이드

등록월 20

별크 + 클러 115,630원 + 104물 □
2위 별크 102,890원 + 142물 □
해외구매 230,600원 + 1물 □



30 인텔

인텔(소켓 최대 클럭 PCIe5.0, 4 인텔 UHC 시네벤치 별도)

기획전
관련기사

사용기

등록월 20

1 인텔 코어i5-13세대 13400F (랩터레이크) 표준PC

인텔(소켓1700) / 10nm(인텔7) / 6+4코어 / 12+4쓰레드 / 기본 클럭: 2.5GHz / 최대 클럭: 4.6GHz / L2 캐시: 9.5MB / L3 캐시: 20MB / PBP/MTP: 65~148W / PCIe5.0, 4.0 / 메모리 규격: DDR5, DDR4 / 4800, 3200MHz / 내장그래픽: 미탑재 / 기술 지원: 하이퍼스레드 / 출시가: 97날라 (VAT별도) / 시네벤치R23(싱글): 1,666점 / 시네벤치R23(멀티): 8376

기획전 PC방 인기 :
관련기사 역대급 대즈 템 [한눈에]
사용기 효율적인 기

등록월 2023.01. | 상품

2 AMD 라이젠5

AMD(소켓AM5) / 5세대(Zen4) / 5nm / 8코어 / 16쓰레드 / 기본 클럭: 4.2GHz / 최대 클럭: 5.0GHz / L2 캐시: 8MB / L3 캐시: 96MB / TDP: 120W / PCIe5.0 / 메모리 규격: DDR5 / 5200MHz / 내장그래픽: 미탑재 / AMD 라데온 그래픽 / 기술 지원: Ryzen Master, 3D V캐시 / 쿨러: 미포함 / 시네벤치R23(싱글): 1,666점 / 시네벤치R23(멀티): 8376

기획전 PC방 인기 :
관련기사 AMD, 글로벌
사용기 디앤디컴 A

등록월 2023.01. | 상품

가격비교 (426)

검색상품

단종상품

중고장터

리스트 내 검색

결과 내 검색

인기상품순 | 신상품순 | 낮은가격순 | 높은가격순 | 판매점순 | 상품의견 많은순

원 ~ 원 검색

30개

AMD 라이젠7-5세대 7800X3D (라파엘) (정품) 표준PC

AMD(소켓AM5) / 5세대(Zen4) / 5nm / 8코어 / 16쓰레드 / 기본 클럭: 4.2GHz / 최대 클럭: 5.0GHz / L2 캐시: 8MB / L3 캐시: 96MB / TDP: 120W / PCIe5.0 / 메모리 규격: DDR5 / 5200MHz / 내장그래픽: 미탑재 / AMD 라데온 그래픽 / 기술 지원: Ryzen Master, 3D V캐시 / 쿨러: 미포함 / 시네벤치R23(싱글): 1,666점 / 시네벤치R23(멀티): 8376

606,000원 + 272물 □

AMD 라이젠5-5세대 7600 (라파엘) (멀티팩(정품)) 표준PC 이벤트

AMD(소켓AM5) / 5세대(Zen4) / 5nm / 6코어 / 12쓰레드 / 기본 클럭: 3.8GHz / 최대 클럭: 5.1GHz / L2 캐시: 6MB / L3 캐시: 32MB / TDP: 65W / PPT: 88W / PCIe5.0 / 메모리 규격: DDR5 / 5200MHz / 내장그래픽: 탑재 / AMD 라데온 그래픽 / 쿨러: Wraith Stealth 포함

254,500원 + 289물 □

인텔 코어i5-13세대 13400F (랩터레이크) 표준PC

인텔(소켓1700) / 10nm(인텔7) / 6+4코어 / 12+4쓰레드 / 기본 클럭: 2.5GHz / 최대 클럭: 4.6GHz / L2 캐시: 9.5MB / L3 캐시: 20MB / PBP/MTP: 65~148W /

Accuracy Metrics for Offline Evaluation

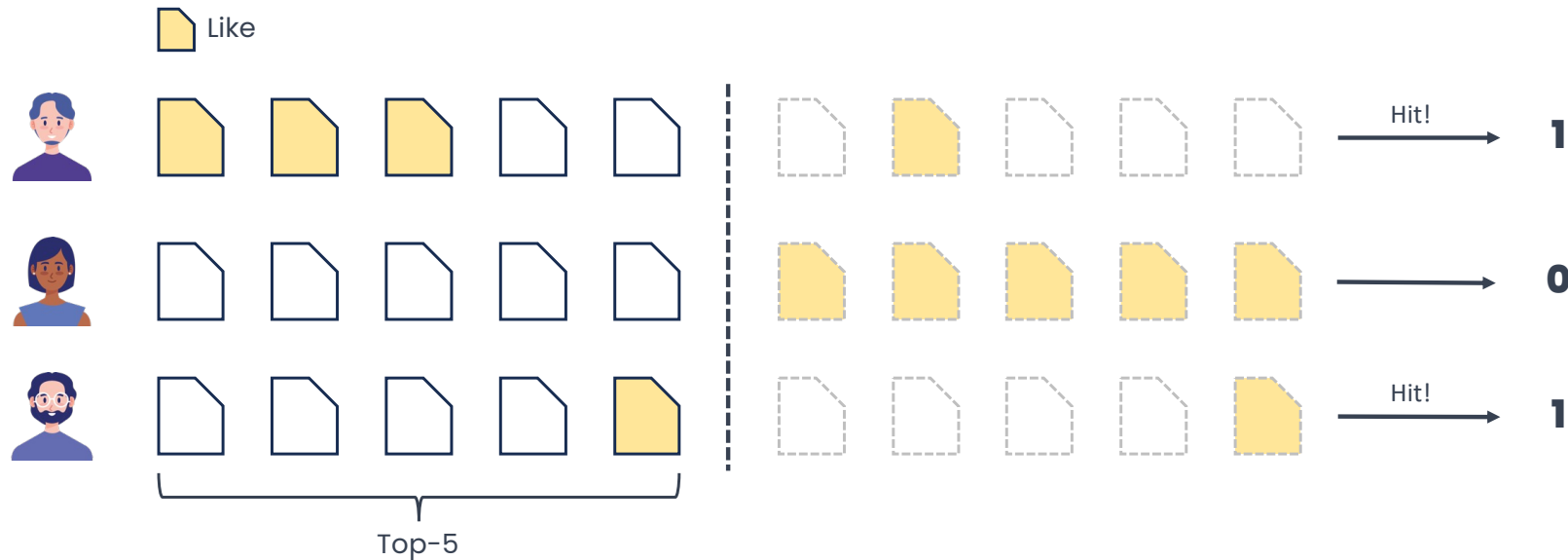
- **Evaluating Ranking List**

- In practice, recommender systems generate a **ranked list of items** for a user, from which the top-k items are recommended
- It is preferable for highly rated items to be ranked higher than those with lower ratings
- The aforementioned measures, designed to evaluate the prediction accuracy of the actual rating value, do not account for this importance

Accuracy Metrics for Offline Evaluation

- **Hit Rate@ k**

- The hit rate at k ($HR@k$) is the simplest evaluation metric, providing a score when there are items the user would like among the top- k recommended items



$$\text{hit rate} = \frac{\# \text{ of hit users}}{\# \text{ of users}}$$

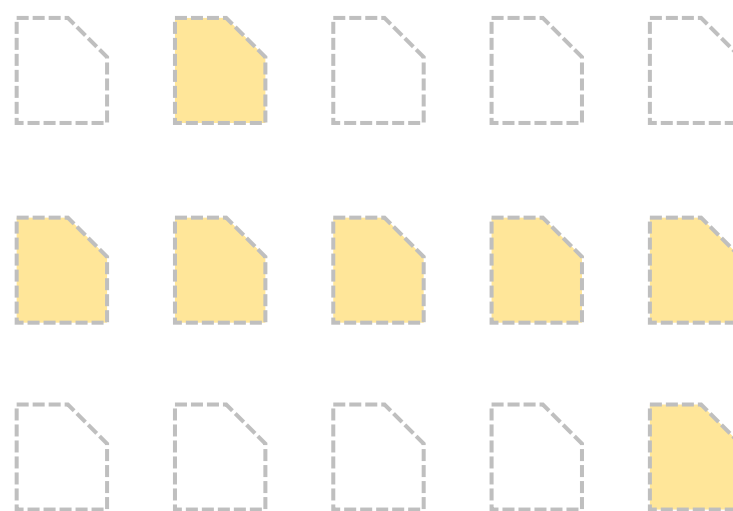
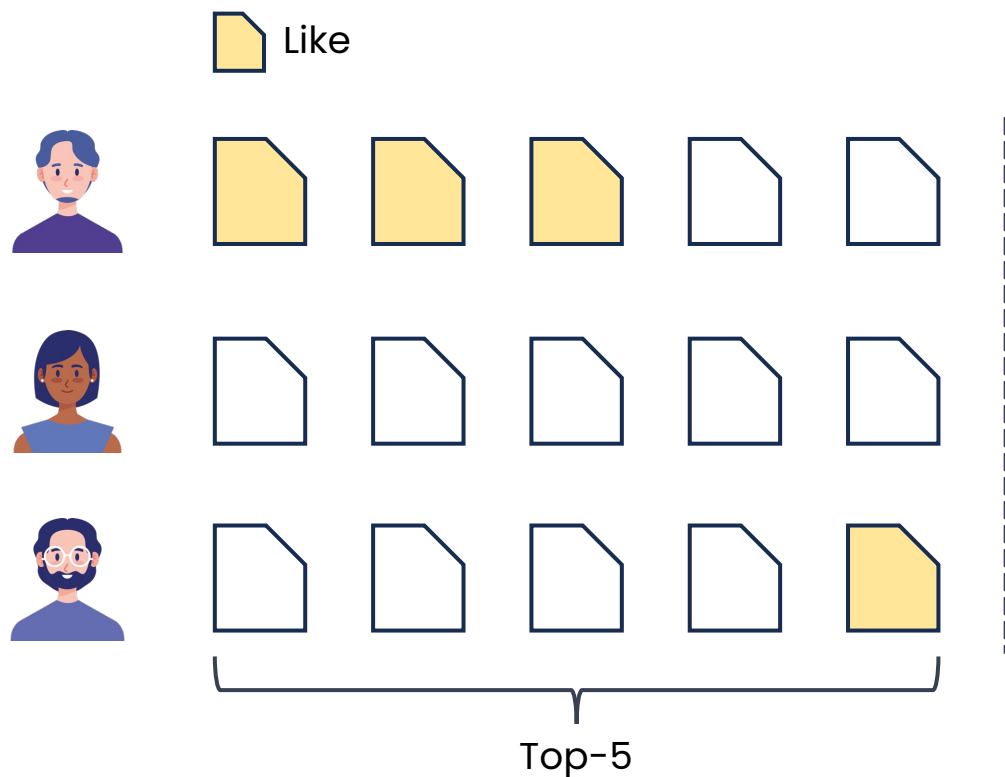
Accuracy Metrics for Offline Evaluation

- **Precision@ k and Recall@ k**

		Model Prediction	
		True	False
Actual Label	True	True Positive (TP)	False Negative (FN)
	False	False Positive (FP)	True Negative (TN)

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$



$$\text{Precision} = \frac{3}{5} = 0.6$$
$$\text{Recall} = \frac{3}{4} = 0.75$$

$$\text{Precision} = \frac{0}{5} = 0.0$$
$$\text{Recall} = \frac{0}{5} = 0.0$$

$$\text{Precision} = \frac{1}{5} = 0.2$$
$$\text{Recall} = \frac{1}{2} = 0.5$$

Accuracy Metrics for Offline Evaluation

- **Average Precision@ k**

- Average Precision (AP) is an evaluation metric that considers both the ranking of recommended items and precision
- The higher the correct items are positioned in the top rankings, the higher the score

 Like



$$AP@5 = \left(\frac{1}{1} + \frac{2}{2} + \frac{3}{3} + \frac{3}{4} + \frac{3}{5} \right) / 5 = 0.87$$



$$AP@5 = \left(\frac{0}{1} + \frac{0}{2} + \frac{1}{3} + \frac{2}{4} + \frac{3}{5} \right) / 5 = 0.29$$

Accuracy Metrics for Offline Evaluation

- **Mean Average Precision@ k**

- Mean Average Precision (MAP) is the average of AP scores across all users

$$MAP@k = \frac{\sum_{u \in U} AP@k(u)}{|U|}$$

 Like



$$AP@5 = \left(\frac{1}{1} + \frac{2}{2} + \frac{3}{3} + \frac{3}{4} + \frac{3}{5} \right) / 5 = 0.87$$



$$AP@5 = \left(\frac{0}{1} + \frac{0}{2} + \frac{1}{3} + \frac{2}{4} + \frac{3}{5} \right) / 5 = 0.29$$

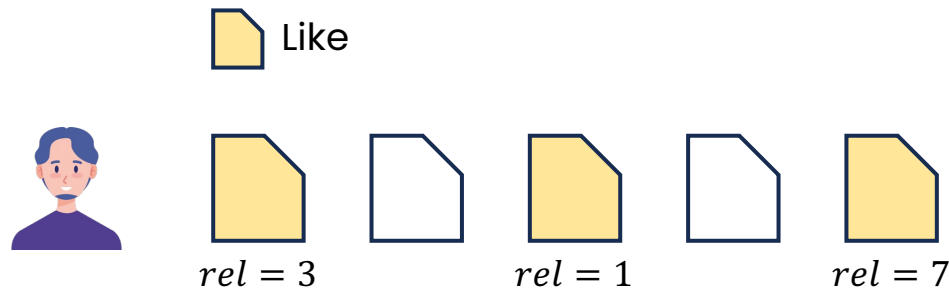
$$MAP@5 = (0.87 + 0.29) / 2$$

Accuracy Metrics for Offline Evaluation

- **Discounted Cumulative Gain@k (DCG@k)**

- Discounted Cumulative Gain (DCG) is an evaluation metric that incorporates the weighting based on the order of recommended items and the utility (relevance) users derive from these items

$$DCG_k = \sum_{i=1}^k \frac{rel_i}{\log_2(i+1)}$$



$$DCG@5 = \frac{3}{\log_2(1+1)} + \frac{0}{\log_2(2+1)} + \frac{1}{\log_2(3+1)} + \frac{0}{\log_2(4+1)} + \frac{7}{\log_2(5+1)} = 6.21$$

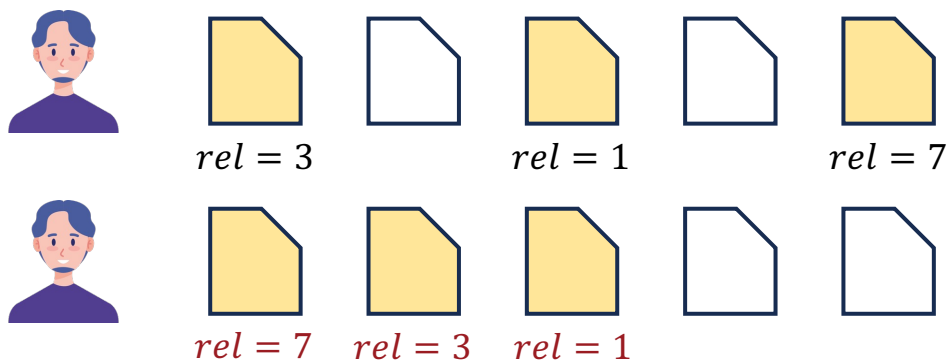
Accuracy Metrics for Offline Evaluation

- **Normalized Discounted Cumulative Gain@k (NDCG@k)**

- Normalized Discounted Cumulative Gain (NDCG) provides normalized DCG values to make it easier to compare the performance of different recommender systems

$$NDCG_k = \frac{DCG_k}{IDCG_k}$$

- Ideal Discounted Cumulative Gain (IDCG) is the upper bound of DCG, calculated by assuming optimal recommendation results



$$DCG@5 = \frac{3}{\log_2(1+1)} + \frac{0}{\log_2(2+1)} + \frac{1}{\log_2(3+1)} + \frac{0}{\log_2(4+1)} + \frac{7}{\log_2(5+1)} = 6.21$$

$$IDCG@5 = \frac{7}{\log_2(1+1)} + \frac{3}{\log_2(2+1)} + \frac{1}{\log_2(3+1)} + \frac{0}{\log_2(4+1)} + \frac{0}{\log_2(5+1)} = 9.39$$

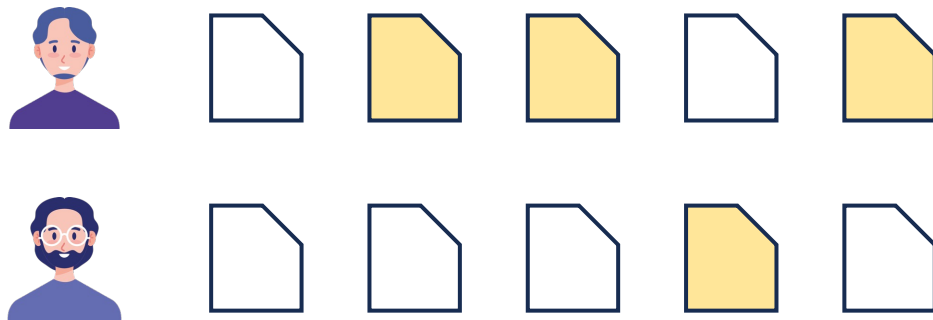
$$NDCG@5 = \frac{DCG@5}{IDCG@5} = \frac{6.21}{9.39} = 0.66$$

Accuracy Metrics for Offline Evaluation

- **Mean Reciprocal Rank (MRR)**

- Mean Reciprocal Rank (MRR) calculates the average of the reciprocals of the positions where the first relevant item appears
- MRR measures the quality of recommender systems by showing the most relevant items to users first

$$MRR = \frac{1}{|U|} \sum_{u \in U} \frac{1}{rank_i}$$



$$MRR = \frac{1}{2} \left(\frac{1}{2} + \frac{1}{4} \right) = 0.375$$

Accuracy Metrics for Offline Evaluation

- Examples of Evaluation Metrics

Algorithm	MovieLoverData		UserItemData	
	MSE	MAE	MSE	MAE
Naive Bayes	0.0346 $\pm$ 0.0002	0.1665 $\pm$ 0.0007	0.0150 $\pm$ 0.0001	0.0990 $\pm$ 0.0005
LR	N/A	N/A	0.0002 $\pm$ 0.0001	0.0105 $\pm$ 0.0017
LR-U	N/A	N/A	0.0070 $\pm$ 0.0000	0.0667 $\pm$ 0.0002
LR-I	N/A	N/A	0.0065 $\pm$ 0.0000	0.0639 $\pm$ 0.0001
1BITMC	0.0520 $\pm$ 0.0003	0.1724 $\pm$ 0.0006	0.0119 $\pm$ 0.0000	0.0881 $\pm$ 0.0002

Table 1: Estimation accuracy of propensity score matrix (average $\pm$ standard deviation across 10 experimental repeats).

Table 2: Overall performance of our method and baseline models on AUC and Logloss metrics. FMCG, FA, GM, EL represents four industry categories of items (fast-moving consumer goods, fashion, general merchandise, and electronics). The best results within groups are with underlines, the best results of all methods are in bold.

Group	Method	overall		FMCG		FA		GM		EL	
		AUC	Logloss	AUC	Logloss	AUC	Logloss	AUC	Logloss	AUC	Logloss
I	SVD++	0.6539	0.6689	0.6507	0.6412	0.6565	0.6764	0.6474	0.6716	0.6487	0.6657
	YoutubeNet	0.6558	0.6746	0.6545	0.6447	0.6584	0.6804	0.6483	0.6825	0.6491	0.6775
	PNN	<u>0.6564</u>	<u>0.6650</u>	<u>0.6552</u>	<u>0.6351</u>	<u>0.6590</u>	<u>0.6726</u>	<u>0.6486</u>	<u>0.6688</u>	<u>0.6496</u>	<u>0.6640</u>

Table 2: The overall comparison. $\star$ indicates a statistically significant level p -value <0.05 comparing DG-ENN with the best baseline (indicated by underlined numbers).

Dataset	Alipay		Tmall		Alimama	
Model	AUC	Logloss	AUC	Logloss	AUC	Logloss
LR	0.8196	0.2276	0.8760	0.1991	0.7207	0.2693
FM	0.8498	0.2175	0.9026	0.1831	0.7396	0.2668
AutoInt+	0.8631	0.2147	0.9181	0.1730	0.7499	0.2611
DeepFM	0.8648	0.2084	0.9155	0.1774	0.7653	0.2581
PNN	<u>0.8756</u>	<u>0.2020</u>	<u>0.9261</u>	<u>0.1650</u>	<u>0.7758</u>	<u>0.2534</u>
DIN	0.8649	0.2081	0.9169	0.1761	0.7644	0.2584
DIEN	0.8731	0.2037	0.9235	0.1684	0.7710	0.2554
GIN	0.8645	0.2093	0.9194	0.1716	0.7621	0.2595
FiGNN	0.8632	0.2121	0.9180	0.1753	0.7438	0.2635
DG-ENN	0.9216*	0.1674*	0.9501*	0.1399*	0.8443*	0.2254*

Accuracy Metrics for Offline Evaluation

- Examples of Evaluation Metrics

Table 3: Recommendation performance of our ARD with SASRec and seven baselines on four datasets. Notice that the best results and the second best results are marked in bold and underlined, respectively.

Dataset	Metric	BPR	FPMC	TransRec	GRU4Rec+	Caser	BERT4Rec	SASRec	ARD
Music	Rec@10	0.3769	0.4087	0.5406	0.4446	0.5180	0.5467	<u>0.5526</u>	0.5890
	NDCG@10	0.2217	0.2505	0.3813	0.2980	0.3341	0.3630	<u>0.3888</u>	0.4184
Tmall	Rec@10	0.4272	0.4675	0.5666	0.5561	0.5525	<u>0.6155</u>	0.6121	0.6459
	NDCG@10	0.2514	0.2862	0.3791	0.3791	0.3496	<u>0.4052</u>	0.4046	0.4373
Baby	Rec@10	0.4145	0.4307	0.4834	0.4232	0.4496	<u>0.5137</u>	0.5092	0.5265
	NDCG@10	0.2200	0.2426	0.2874	0.2404	0.2700	<u>0.3131</u>	0.2854	0.3150

Dataset	Model	NDCG@100	Recall@20	Recall@50
ML20M (0.002)	EASE	0.420	0.391	0.521
	EASE - IPW	0.424	0.395	0.526
	ARM	0.419	0.389	0.520
	R-LARM	0.425	0.395	0.527
	R-LARM(NN)	0.426	0.396	0.529
NFLX (0.001)	EASE	0.393	0.362	0.445
	EASE - IPW	0.398	0.366	0.449
	ARM	0.393	0.361	0.444
	R-LARM	0.397	0.365	0.448
	R-LARM(NN)	0.398	0.365	0.449
MSD (0.001)	EASE	0.390	0.334	0.428
	EASE - IPW	0.391	0.334	0.429
	ARM	0.390	0.335	0.428
	R-LARM	0.390	0.335	0.428
	R-LARM(NN)	0.391	0.334	0.429

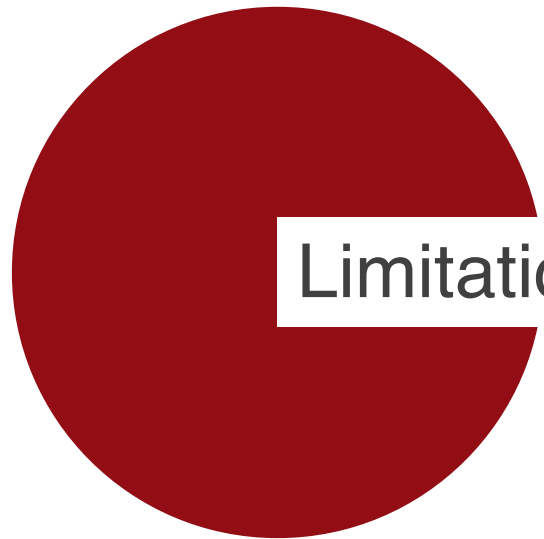
Table 2: Average results on three public datasets by various methods. The error bars [25] on the three datasets are shown along with the names of the datasets. Both EASE-IPW and R-LARM variants perform significantly better compared to the baseline EASE model. Between EASE-IPW and R-LARM models, the performance is almost comparable. On ML20M dataset, we observe R-LARM(NN) performing significantly better compared to EASE-IPW. The ARM models merely perform comparable to the baseline EASE models with no improvements. A direct comparison with many other methods can be seen in [25] which shows that the performance on all these datasets is near or at the state-of-the-art.

Accuracy Metrics for Offline Evaluation

- Examples of Evaluation Metrics

Table 2: Model results and comparison to three benchmark algorithms

Model	Recall@5	Recall@10	Recall@20	MAP@5	MAP@10	MAP@20	MRR@5	MRR@10	MRR@20	NDCG@5	NDCG@10	NDCG@20
Steam												
Popularity	0.0012	0.0019	0.0019	0.0006	0.0007	0.0007	0.0010	0.0012	0.0012	0.0008	0.0011	0.0011
BPR	0.4254	0.4523	0.4861	0.3328	0.3369	0.3397	0.3840	0.3879	0.3903	0.3693	0.3788	0.3882
DAM	0.6561	0.9216	0.9660	0.3951	0.4426	0.4472	0.4395	0.4716	0.4742	0.4740	0.5691	0.5817
BGCN	0.6078	0.8062	0.9374	0.4322	0.4658	0.4774	0.4735	0.4962	0.5046	0.4886	0.5576	0.5935
BRUCE	0.8799*	0.9568*	0.9806*	0.6711*	0.6862*	0.6901*	0.7065*	0.7136*	0.7160*	0.7343*	0.7622*	0.7701*
Improv. (%)	34.11	3.82	1.51	55.28	47.32	44.55	49.21	43.81	41.89	50.29	33.93	29.76
Youshu												
Popularity	0.0027	0.0031	0.0054	0.0017	0.0016	0.0018	0.0039	0.0042	0.0048	0.0024	0.0025	0.0034
BPR	0.0764	0.1034	0.1532	0.0430	0.0435	0.0473	0.0942	0.1035	0.1095	0.0638	0.0729	0.0890
DAM	0.1098	0.1460	0.2000	0.0618	0.0644	0.0681	0.1167	0.1261	0.1321	0.0873	0.0996	0.1162
BGCN	0.0942	0.1401	0.1992	0.0614	0.0630	0.0668	0.1235	0.1334	0.1394	0.0880	0.1002	0.1170
BRUCE	0.1344*	0.1893*	0.2624*	0.0743*	0.0769*	0.0822*	0.1465*	0.1602*	0.1671*	0.1068*	0.1249*	0.1471*
Improv. (%)	22.40	29.66	31.20	20.23	19.41	20.70	18.62	20.09	19.87	21.36	24.65	25.73
NetEase												
Popularity	0.0004	0.0007	0.0013	0.0003	0.0003	0.0003	0.0006	0.0007	0.0008	0.0004	0.0005	0.0007
BPR	0.0131	0.0212	0.0363	0.0064	0.0071	0.0081	0.0172	0.0204	0.0230	0.0107	0.0139	0.0188
DAM	0.0099	0.0163	0.0293	0.0048	0.0055	0.0064	0.0118	0.0138	0.0158	0.0077	0.0102	0.0142
BGCN	0.0138	0.0244	0.0427	0.0070	0.0080	0.0092	0.0172	0.0207	0.0236	0.0115	0.0154	0.0210
BRUCE	0.0183*	0.0296*	0.0489*	0.0092*	0.0103*	0.0117*	0.0218*	0.0258*	0.0289*	0.0145*	0.0189*	0.0251*
Improv. (%)	32.61	21.31	14.52	31.43	28.75	27.17	26.74	24.64	22.46	26.09	22.73	19.52



Limitations of Evaluation Measures

Limitations of Evaluation Measures

- **Evaluation cannot be fully achieved using a single evaluation metric**
 - Relying on a single evaluation metric is insufficient when assessing the performance of a recommendation system
 - Instead, using multiple evaluation metrics together is appropriate for a comprehensive assessment of the model's performance, tailored to specific evaluation goals

Limitations of Evaluation Measures

- **Data is highly imbalanced and sparse**
 - User-item interaction data follow the long-tailed distribution, where a few items (the head or major items) receive the majority interactions, many others (the long tail or minor items) receive relatively few interactions

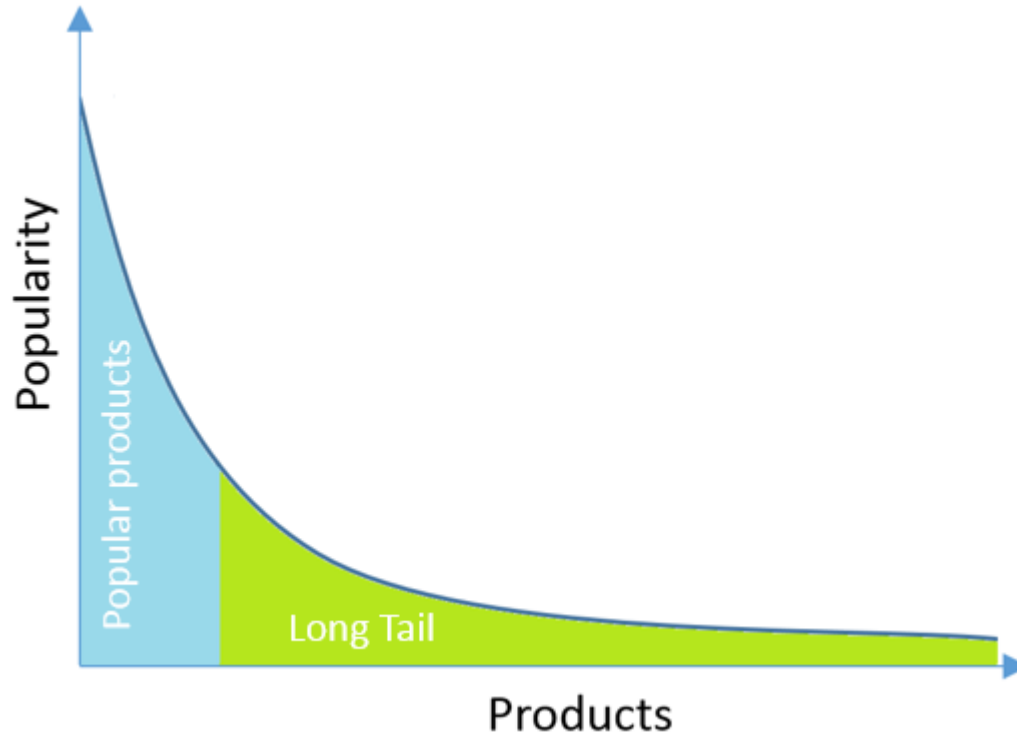


Table 1: Dataset statistics.

Dataset	#user	#item	#rating	rating density	#social relation	social density
Epinions	11,500	7,596	283,319	0.32%	275,117	0.21%
Lthing	15,039	14,957	529,992	0.24%	44,710	0.02%

Table 2: Statistics of the Datasets.

Datasets	#Users	#Items	#Interactions	Sparsity
Gowalla	29,858	40,981	1,027,464	99.92%
Amazon-Books	52,642	91,598	2,984,108	99.94%
Yelp2018	31,668	38,048	1,561,406	99.87%
Alibaba-iFashion	300,000	81,614	1,607,813	99.99%

Limitations of Evaluation Measures

- **Data is highly imbalanced and sparse**

- Popular items are more likely to receive a higher number of ratings from users
- As a result, recommendation models tend to recommend popular items more frequently
- Therefore, the following factors should also be considered when evaluating recommendation models:
 - **Coverage**: Long-tail items can sometimes have significant value. Therefore, it is important to recommend not only specific items but also a diverse range of items from the entire catalog to users
 - **Novelty**: It is crucial to continuously recommend new items that users have not seen before, rather than merely suggesting items they have already been recommended in the past
 - **Diversity**: The recommendation list should not consist solely of similar items. It is important to include a diverse range of items in the recommendations
 - **Serendipity**: By recommending items that users might not have considered, you can provide them serendipitous enjoyment, which can enhance their satisfaction with the service
 - **Scalability**: Due to the vast number of items and users, a recommendation system needs to be capable of handling large-scale data effectively. Recommendations should be generated quickly and efficiently to manage the massive volume of users and items